Lactococcus lactis, a promising and efficient factory for the bioproduction of functional soluble and membrane proteins

Auteurs : Escrihuela Jimenez I., Labbé C., Amara M.M., Azizi M., ***Frelet-Barrand A***

Affiliation : Université Marie et Louis Pasteur, CNRS, institut FEMTO-ST, F-25000 Besançon

Lactococcus lactis, a gram-positive bacterium considered the model lactic acid bacteria, emerged at the beginning of the last century as a good alternative for protein expression and characterizations. Already largely used in industry for production of fermented foods and consumed by humans for hundreds of years, this "generally recognized as safe" (GRAS) microorganism can express functional soluble and membrane proteins (MPs), including therapeutic proteins and bacterial and viral antigens for vaccination even at the industrial scale. *L. lactis* is an interesting alternative host because of its moderate proteolytic activity, the absence of inclusion-body formation and of endotoxin production, and its efficient targeting of the MPs into a single plasma membrane. This system also allows properly folded expression of proteins in their native oligomeric form.

Among the different gene expression systems, the tightly regulated nisin-controlled gene expression (NICE) system is the most broadly used. This system is based on genes involved in the biosynthesis and regulation of the antimicrobial peptide nisin. Using the NICE system, until 2022, 115 MPs from prokaryotic or eukaryotic origins, with diverse topologies and sizes, have been successfully expressed allowing their functional and/or structural characterizations.

Over the last two decades, *L. lactis* emerged and proved to be an alternative and promising expression system for both soluble and membrane proteins compared to other systems. This attractive cell factory will enrich the knowledge of proteins, including MPs in their functional and structural states, and bring about the development of further biotechnological and biotherapeutic applications in the future.